Experiment no 7

Implementing Part of Speech (POS) tagging with (NLTK/spacy/ Stanford) in Python NLTK

```
import nltk

# Download the required NLTK resources
nltk.download("punkt")
nltk.download("punkt_tab")
nltk.download("averaged_perceptron_tagger")
nltk.download("averaged_perceptron_tagger_eng")

def pos_tag_sentence(sentence):
    # Tokenize the sentence
    tokens = nltk.word_tokenize(sentence)

    # Perform POS tagging
    tagged_words = nltk.pos_tag(tokens)

    return tagged_words


# Multiple input sentences for exploration
sentences = [
    "Book that flight.",
    "Hand me that book.",
    "Does that flight serve dinner?",
    "I read a book.",
    "Please book a ticket.",
    "The quick brown fox jumps over the lazy dog.",
    "She is running in the park.",
    "The children are playing football.",
    "He can swim.",
    "I can a can.",
    "Time flies like an arrow.",
    "Fruit flies like a banana.",
    "The students study NLP.",
    "The beautiful flowers bloom in spring.",
    "She writes a letter and he reads a book."
]

# Display POS tags for every sentence
for i, sentence in enumerate(sentences, start=1):
    print(f"\nInput {i}: {sentence}")
    print("Output:")

    tagged_words = pos_tag_sentence(sentence)
```

```
    for word, tag in tagged_words:
        print(f"{word} / {tag}", end="  ")

    print()
```

Output

Input 1: Book that flight.
Output:
Book / NNP that / WDT flight / NN . / .

Input 2: Hand me that book.
Output:
Hand / NNP me / PRP that / IN book / NN . / .

Input 3: Does that flight serve dinner?
Output:
Does / VBZ that / IN flight / NN serve / NN dinner / NN ? / .

Input 4: I read a book.
Output:
I / PRP read / VBP a / DT book / NN . / .

Input 5: Please book a ticket.
Output:
Please / NNP book / NN a / DT ticket / NN . / .

Input 6: The quick brown fox jumps over the lazy dog.
Output:
The / DT quick / JJ brown / NN fox / NN jumps / VBZ over / IN the / DT lazy / JJ dog / NN . / .

Input 7: She is running in the park.
Output:
She / PRP is / VBZ running / VBG in / IN the / DT park / NN . / .

Input 8: The children are playing football.
Output:
The / DT children / NNS are / VBP playing / VBG football / NN . / .

Input 9: He can swim.
Output:
He / PRP can / MD swim / VB . / .

Input 10: I can a can.
Output:
I / PRP can / MD a / DT can / MD . / .

Input 11: Time flies like an arrow.

Output:
Time / NNP flies / NNS like / IN an / DT arrow / NN . / .

Input 12: Fruit flies like a banana.
Output:
Fruit / NNP flies / VBZ like / IN a / DT banana / NN . / .

Input 13: The students study NLP.
Output:
The / DT students / NNS study / VBP NLP / NNP . / .

Input 14: The beautiful flowers bloom in spring.
Output:
The / DT beautiful / JJ flowers / NNS bloom / VBP in / IN spring / NN . / .

Input 15: She writes a letter and he reads a book.
Output:
She / PRP writes / VBZ a / DT letter / NN and / CC he / PRP reads / VBZ a / DT book / NN . / .

spaCy

```
!pip install nltk spacy
!python -m spacy download en_core_web_sm
import spacy

nlp = spacy.load("en_core_web_sm")

sentences = [
    "Book that flight.",
    "Hand me that book.",
    "Does that flight serve dinner?",
    "I read a book.",
    "Please book a ticket.",
    "The quick brown fox jumps over the lazy dog.",
    "She is running in the park.",
    "The children are playing football.",
    "He can swim.",
    "The students study NLP.",
    "The beautiful flowers bloom in spring.",
    "She writes a letter and he reads a book.",
    "Time flies like an arrow.",
    "Fruit flies like a banana.",
    "Python is a powerful programming language."
]

for sentence in sentences:
    doc = nlp(sentence)
```

```
    print("\nInput:", sentence)
    print("Output:", end=" ")

    for token in doc:
        print(f"{token.text} / {token.tag_}", end="   ")

    print()
```

Output

```
Input: Book that flight.
Output: Book / NN   that / DT   flight / NN   . / .

Input: Hand me that book.
Output: Hand / VB   me / PRP   that / DT   book / NN   . / .

Input: Does that flight serve dinner?
Output: Does / VBZ   that / DT   flight / NN   serve / VB   dinner / NN   ? / .

Input: I read a book.
Output: I / PRP   read / VBD   a / DT   book / NN   . / .

Input: Please book a ticket.
Output: Please / UH   book / VB   a / DT   ticket / NN   . / .

Input: The quick brown fox jumps over the lazy dog.
Output: The / DT   quick / JJ   brown / JJ   fox / NN   jumps / VBZ   over / IN   the / DT   lazy / JJ
dog / NN   . / .

Input: She is running in the park.
Output: She / PRP   is / VBZ   running / VBG   in / IN   the / DT   park / NN   . / .

Input: The children are playing football.
Output: The / DT   children / NNS   are / VBP   playing / VBG   football / NN   . / .

Input: He can swim.
Output: He / PRP   can / MD   swim / VB   . / .

Input: The students study NLP.
Output: The / DT   students / NNS   study / VBP   NLP / NNP   . / .

Input: The beautiful flowers bloom in spring.
Output: The / DT   beautiful / JJ   flowers / NNS   bloom / VBP   in / IN   spring / NN   . / .

Input: She writes a letter and he reads a book.
Output: She / PRP   writes / VBZ   a / DT   letter / NN   and / CC   he / PRP   reads / VBZ   a / DT
book / NN   . / .
```

Input: Time flies like an arrow.
Output: Time / NN flies / VBZ like / IN an / DT arrow / NN . / .

Input: Fruit flies like a banana.
Output: Fruit / NN flies / VBZ like / IN a / DT banana / NN . / .

Input: Python is a powerful programming language.
Output: Python / NNP is / VBZ a / DT powerful / JJ programming / NN language / NN . / .

Stanza

```
!pip install stanza
import stanza

# Download the English model
stanza.download("en")

# Initialize the English pipeline
nlp = stanza.Pipeline(
    lang="en",
    processors="tokenize,pos",
    verbose=False
)

sentences = [
    "Book that flight.",
    "Hand me that book.",
    "Does that flight serve dinner?",
    "I read a book.",
    "Please book a ticket.",
    "The quick brown fox jumps over the lazy dog.",
    "She is running in the park.",
    "The children are playing football.",
    "He can swim.",
    "The students study NLP.",
    "The beautiful flowers bloom in spring.",
    "She writes a letter and he reads a book.",
    "Time flies like an arrow.",
    "Fruit flies like a banana.",
    "Python is a powerful programming language."
]

for sentence in sentences:
    doc = nlp(sentence)

    print("\nInput:", sentence)
    print("Output:", end=" ")
```

```
    for sent in doc.sentences:
        for word in sent.words:
            print(f"{word.text} / {word.xpos}", end="   ")

    print()
```

Output

Input: Book that flight.
Output: Book / VB that / DT flight / NN . / .

Input: Hand me that book.
Output: Hand / VB me / PRP that / DT book / NN . / .

Input: Does that flight serve dinner?
Output: Does / VBZ that / DT flight / NN serve / VB dinner / NN ? / .

Input: I read a book.
Output: I / PRP read / VBD a / DT book / NN . / .

Input: Please book a ticket.
Output: Please / UH book / VB a / DT ticket / NN . / .

Input: The quick brown fox jumps over the lazy dog.
Output: The / DT quick / JJ brown / JJ fox / NN jumps / VBZ over / IN the / DT lazy / JJ dog / NN . / .

Input: She is running in the park.
Output: She / PRP is / VBZ running / VBG in / IN the / DT park / NN . / .

Input: The children are playing football.
Output: The / DT children / NNS are / VBP playing / VBG football / NN . / .

Input: He can swim.
Output: He / PRP can / MD swim / VB . / .

Input: The students study NLP.
Output: The / DT students / NNS study / VBP NLP / NN . / .

Input: The beautiful flowers bloom in spring.
Output: The / DT beautiful / JJ flowers / NNS bloom / VBP in / IN spring / NN . / .

Input: She writes a letter and he reads a book.
Output: She / PRP writes / VBZ a / DT letter / NN and / CC he / PRP reads / VBZ a / DT book / NN . / .

Input: Time flies like an arrow.
Output: Time / NN flies / VBZ like / IN an / DT arrow / NN . / .

Input: Fruit flies like a banana.
Output: Fruit / NN flies / VBZ like / IN a / DT banana / NN . / .

Input: Python is a powerful programming language.
Output: Python / NNP is / VBZ a / DT powerful / JJ programming / NN language / NN . / .